

Q-Orbit: Fail-Closed Finite-Key Screening and Conceptual Engineering Architecture for Satellite QKD

A Controlled Preliminary Theoretical Research Paper with Illustrated System Concepts

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Release: **PRIVATE-BLOCKED** | Phase 1: **COMPLETE**

Gate: **CONDITIONAL PASS - READY FOR PHASE 2 WITH DISCLOSED LIMITATIONS**

INTERIM THEORETICAL SUBMISSION — NOT PHYSICALLY VALIDATED

Prototype: BUILT - Q-Orbit Theoretical Analysis Console V0.17 (software/research only). Key release: QUARANTINED / ZERO RELEASED.

Scientific Manuscript V1.0-RC2: COMPLETE. Physical Characterization: NOT EXECUTED.

Website: IN PROGRESS - exhibition/research interface; excluded from the scientific validation record.

Edition note. This illustrated academic edition integrates controlled drawings QO-DWG-101 through QO-DWG-104 into the complete V1.0-RC2 preliminary manuscript. The figures clarify functional allocation and trust boundaries only. No scientific conclusion, canonical numerical value, proof status, or physical-validation status has been changed.

Claim control. Numerical reproduction: PASS-THEORETICAL.

Physical gates: BLOCKED/DEFERRED. Unspecified design values: TBD.

Abstract

Satellite quantum key distribution is constrained by short optical-access windows and finite detection blocks. This paper reports a controlled theoretical analysis for Q-Orbit, a single satellite-to-ground optical-link key-replenishment concept. The study freezes an efficient-BB84 weak-coherent-pulse profile with three intensities including vacuum, reproduces a reference finite-key fixture, re-optimizes the integration half-window for every evaluated point, and evaluates both local model response and a deterministic two-parameter screen. The reference fixture is reproduced at a 102 s half-window with a signed finite-key margin of **41,338.62418456675 bits** and a floored candidate key of **41,338 bits**. A 41×41 screen over controlled extraneous-count and intrinsic-QBER ranges yields **568 positive and 1,113 nonpositive points**, corresponding to a grid fraction of **0.33789411064842356 (33.789411%)** and a **median signed margin of -2,624.947 bits**. The grid is not a probability distribution. Zero of the 15 mandated effects are MAPPED-IN-CURRENT-FIXTURE; five require proof work, two are BLOCKING, and the remainder require security-budget or characterization closure, preventing implementation-security, certification, mission, or procurement claims. The main contribution is an evidence architecture that couples reproducible numerical screening to fail-closed claim controls.

Keywords: satellite QKD; finite-key analysis; decoy-state BB84; device characterization; evidence control; reproducibility.

1. Introduction

Satellite QKD has been studied as a mechanism for extending quantum key establishment beyond the attenuation limits of terrestrial fibre. In low-Earth orbit, however, a ground station observes a satellite for a limited time, so finite-block statistical effects can dominate the key-length calculation [1]. A positive center-case result is therefore not sufficient evidence of a robust mission capability.

Q-Orbit studies one bounded satellite-to-ground topology; the optical link direction remains TBD. The project does not attempt to claim a deployed network, a trusted-relay service, or an arbitrary remote-consumer distribution system. Instead, it asks whether a reproducible finite-key workflow can be connected to explicit system and evidence gates without converting assumptions into unearned security or performance claims.

1.1 Research Question

If only parameters already represented in a frozen Q-Orbit finite-key fixture are varied, how does the optimized signed finite-key margin respond, and which source or detector imperfections still lack a selected proof-compatible numerical map?

1.2 Contribution

- Exact reproduction of a controlled satellite-QKD finite-key fixture (bit-exact verification against hash-verified artifacts).
 - Re-optimization of the integration half-window at every perturbation point.
 - A deterministic two-parameter screen with explicit non-probabilistic interpretation.
 - A proof-to-device mapping that records unmapped effects rather than inventing penalties.
 - A claim-boundary method that separates numerical verification from physical validation and operational security.
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2. Background and Related Work

2.1 Finite-Block Satellite QKD

Sidhu et al. provide a finite-block analysis for weak-coherent-pulse, efficient-BB84 satellite QKD with three intensities and show that system loss, extraneous counts, protocol parameters, and limited pass duration materially affect single-pass secret-key length [1]. Q-Orbit retains this family as a frozen reference profile for deterministic regression and screening; it does not import the source paper's device or mission values as Q-Orbit evidence.

2.2 Decoy-State Assumptions and Source Flaws

Standard decoy-state methods rely on assumptions about pulse statistics. Nahar et al. analyze imperfect phase randomization and non-identically distributed laser pulses, demonstrating that generalized treatment is required when ideal source assumptions do not hold [2]. Xu et al. show that source flaws must be characterized and integrated into the security analysis rather than hidden inside an ideal preparation model [3].

2.3 Characterization-to-Proof Linkage

Tan and Nahar distinguish device-characterization conclusions from the conditions required by a security proof and identify requirements for justified parameter domains and composable reasoning [4]. This distinction motivates Q-Orbit's refusal to replace missing characterization with a convenient scalar penalty.

3. Methods

3.1 Controlled Reference Profile

The analysis retains the V0.6 efficient-BB84 weak-coherent-pulse fixture with one signal intensity and two decoy intensities including vacuum. Expected counts are computed from a modeled reference loss curve. The profile is a software fixture, not a physical device model.

Equation 1. Signed finite-key margin.

$$M = s_{X,0} + s_{X,1} [1 - h_2(\phi_X)] - \lambda_{EC} - 6 \log_2 (21/\varepsilon_s) - \log_2 (2/\varepsilon_c)$$

The candidate key is $\text{floor}(\max(M, 0))$ under the frozen fixture convention. The signed quantity M is retained to identify boundary behavior. A nonpositive value is a model outcome; it is not a measured outage or security-failure probability.

3.2 Window Optimization

For every baseline, perturbation, frontier, and grid point, the model evaluates integer half-windows from 1 to 221 s and selects the window that maximizes signed margin. Smaller half-window breaks an exact tie. The rule avoids silently carrying a baseline-optimal window into a changed count/error regime.

3.3 Local Response

Eight parameters already present in the software contract are varied: additional system loss, detector-efficiency multiplier, source repetition rate, signal-intensity multiplier, weak-decoy multiplier, extraneous-count probability, afterpulse probability, and intrinsic QBER. The declared steps are plus/minus 0.1 dB for additional loss and plus/minus 1% relative for the remaining variables. Because the steps and parameter meanings differ, the resulting rank is local and step-dependent.

3.4 Deterministic Coupled Screen

The screen evaluates 41 extraneous-count values from $1e-7$ to $2e-6$ per pulse and 41 intrinsic QBER values from 0.003 to 0.015, explicitly including the baseline point. The Cartesian product contains 1,681 deterministic points. The ranges are controlled engineering bounds inherited from V0.7, not calibrated distributions.

3.5 Proof-Mapping Rule

Each source or detector effect is assigned a controlled state: mapped stress only, partial scalar stress, unmapped proof required, unmapped model required, unmapped security budget, or alternative not selected. A numerical penalty is applied only when the scalar is explicitly represented. Unmapped effects remain blocking obligations.

3.6 Verification

The implementation verifies controlled-input hashes, exact baseline outputs, state partitioning of the grid, finite numerical values, boundary invariants, and release-state constraints. Twelve regression tests and a 20-check independent package audit pass. An independent V0.13 implementation previously compared 50 locked vectors over 971 metrics with zero open numerical discrepancies.

4. Results

4.1 Baseline Reproduction

Quantity	Result	Evidence
Half-window	102 s	EV-1a (run summary)
Signed margin	41,338.62418456675 bits	EV-1a + EV-2 invariant
Floored candidate key	41,338 bits	EV-1a + EV-2 (floor)
X-basis QBER	0.017422686665352745	EV-1a + recomputation
Phase-error bound φ_X	0.09270161340569935	EV-1a
Single-photon lower bound $s_{X,1}$	183,803.04893680647	EV-1a

4.2 Local Response

The largest absolute local response is associated with additional system loss (−8.216% of the baseline margin per 0.1 dB step), followed by detector efficiency (+3.567% per 1% relative step) and source repetition rate (+1.946% per 1% relative step). These values describe model response around one fixture point.

4.3 Coupled Screen

Statistic	Result	Evidence
Total points	1,681	EV-1b (recomputed)
Positive signed margin	568	EV-1b
Nonpositive signed margin	1,113	EV-1b
Positive grid fraction	33.789411%	EV-1b + EV-2 (568/1681)
Median signed margin	−2,624.946810258186 bits	EV-1b + EV-1a + EV-2
Minimum / maximum signed margin	−3,828.414517626367 / +142,540.7481180454 bits	EV-1b + EV-1a

Critical disclosure (CFR N-26/N-07/N-08): Because the half-window is re-optimized per evaluated point, this partition is an **upper-envelope** quantity relative to any fixed-window evaluation. The positive count at any fixed window is ≤ 568 . The grid fraction is a deterministic screen partition, not a probability, reliability, availability, or mission-success figure.

4.4 Unmapped Device Effects

Effect	Current State	Claim Consequence
Incomplete phase randomization	UNMAPPED-PROOF-REQUIRED	Standard decoy applicability not established
Pulse/intensity correlations	UNMAPPED-PROOF-REQUIRED	IID pulse assumption not established
State-preparation flaws/leakage	UNMAPPED-PROOF-REQUIRED	Aggregate QBER is insufficient
Dead time, recovery, saturation	UNMAPPED-MODEL-REQUIRED	Detector/rate interpretation blocked
Efficiency mismatch	UNMAPPED-PROOF-REQUIRED	Receiver measurement model incomplete
Characterization confidence	UNMAPPED-SECURITY-BUDGET	No certification-to-proof guarantee
Aging/cross-instance memory	UNMAPPED-MODEL-REQUIRED	Persistent operation claim blocked

(Table shows 7 of 15 mandated effects; the full controlled matrix remains in the Phase 1 consolidated package.)

5. Conceptual System Architecture and Engineering Figures

The verified numerical model does not define a flight article or field installation. Figures 1-4 translate the current functional architecture into controlled engineering concepts so that interfaces, evidence gaps, and trust boundaries can be reviewed before physical design. Every figure is **Engineering decision / conceptual - not measured, PRELIMINARY, NOT TO SCALE, and NOT FOR FABRICATION**. No geometric value, supplier, telescope, orbit, parcel, link direction, or hardware performance is released by these drawings.

5.1 Space-Terminal Functional Allocation

Figure 1 identifies a preliminary functional allocation for a conceptual LEO quantum optical terminal. It is an interface-discussion aid only; the optical aperture, pointing assembly, quantum source/detector allocation, onboard processing, thermal design, power system, structure, and all mechanical interfaces remain TBD.

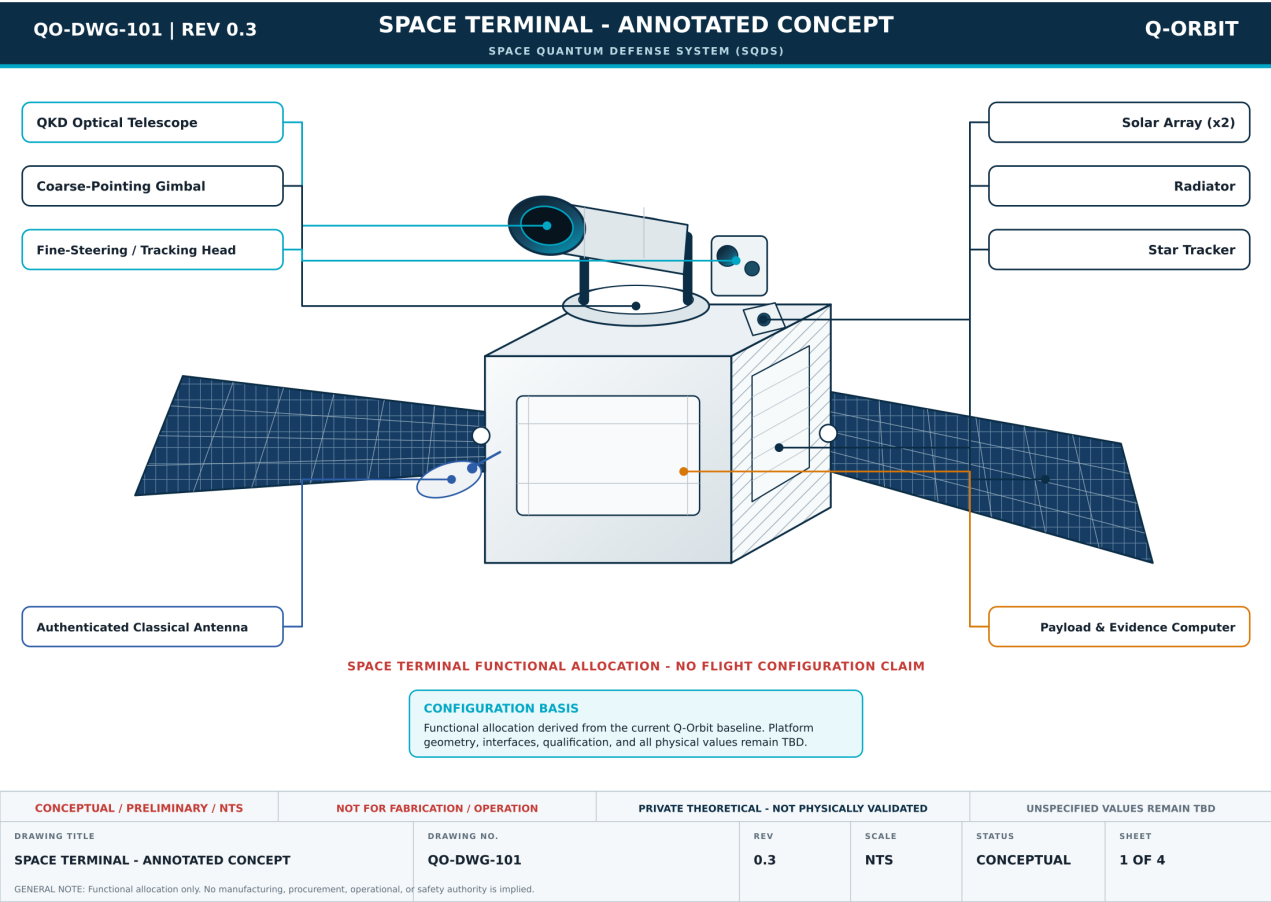


Figure 1: QO-DWG-101 Rev 0.3 - Space-terminal annotated concept. Engineering decision / conceptual - not measured. Functional allocation only; geometry, interfaces, mass, power, thermal performance, and supplier selection are TBD. PRELIMINARY - NOT TO SCALE - NOT FOR FABRICATION - NOT PHYSICALLY VALIDATED.

5.2 Ground-Terminal Functional Allocation

Figure 2 identifies the conceptual ground-terminal functions required to receive or transmit an optical quantum link, subject to the unresolved link-direction decision. The Tabuk reference is a study region only and does not identify an authorized parcel or operating site.

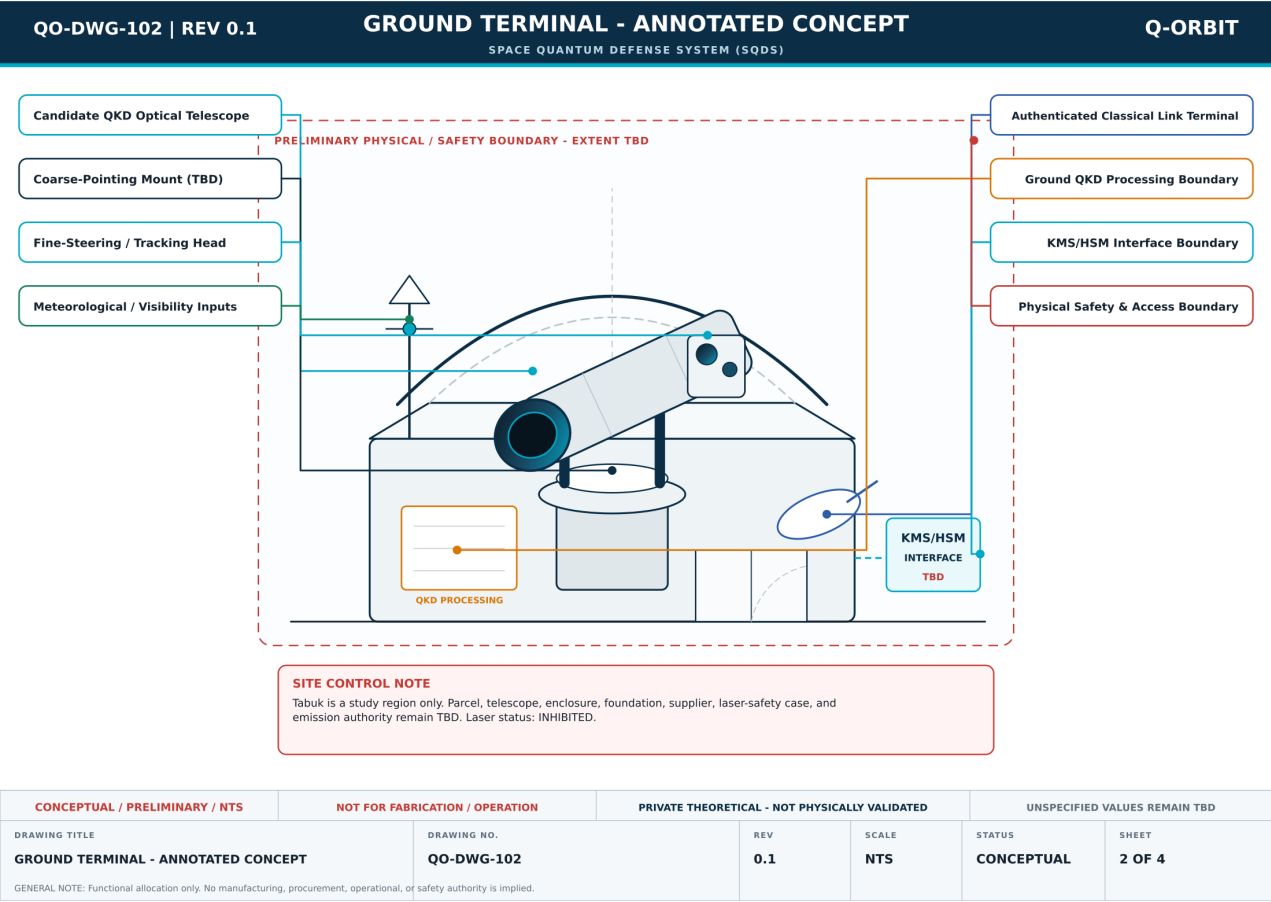


Figure 2: QO-DWG-102 Rev 0.1 - Ground-terminal annotated concept. Engineering decision / conceptual - not measured. Tabuk is a study region only; parcel, telescope, enclosure, foundation, supplier, laser-safety case, emission authority, and environmental qualification remain TBD. Laser state: INHIBITED. PRELIMINARY - NOT TO SCALE - NOT FOR CONSTRUCTION - NOT PHYSICALLY VALIDATED.

5.3 Orthographic Allocation View

Figure 3 presents orthographic views to make the internal functional allocation reviewable from multiple directions. The absence of released dimensions is deliberate: no physical envelope has been validated or approved.

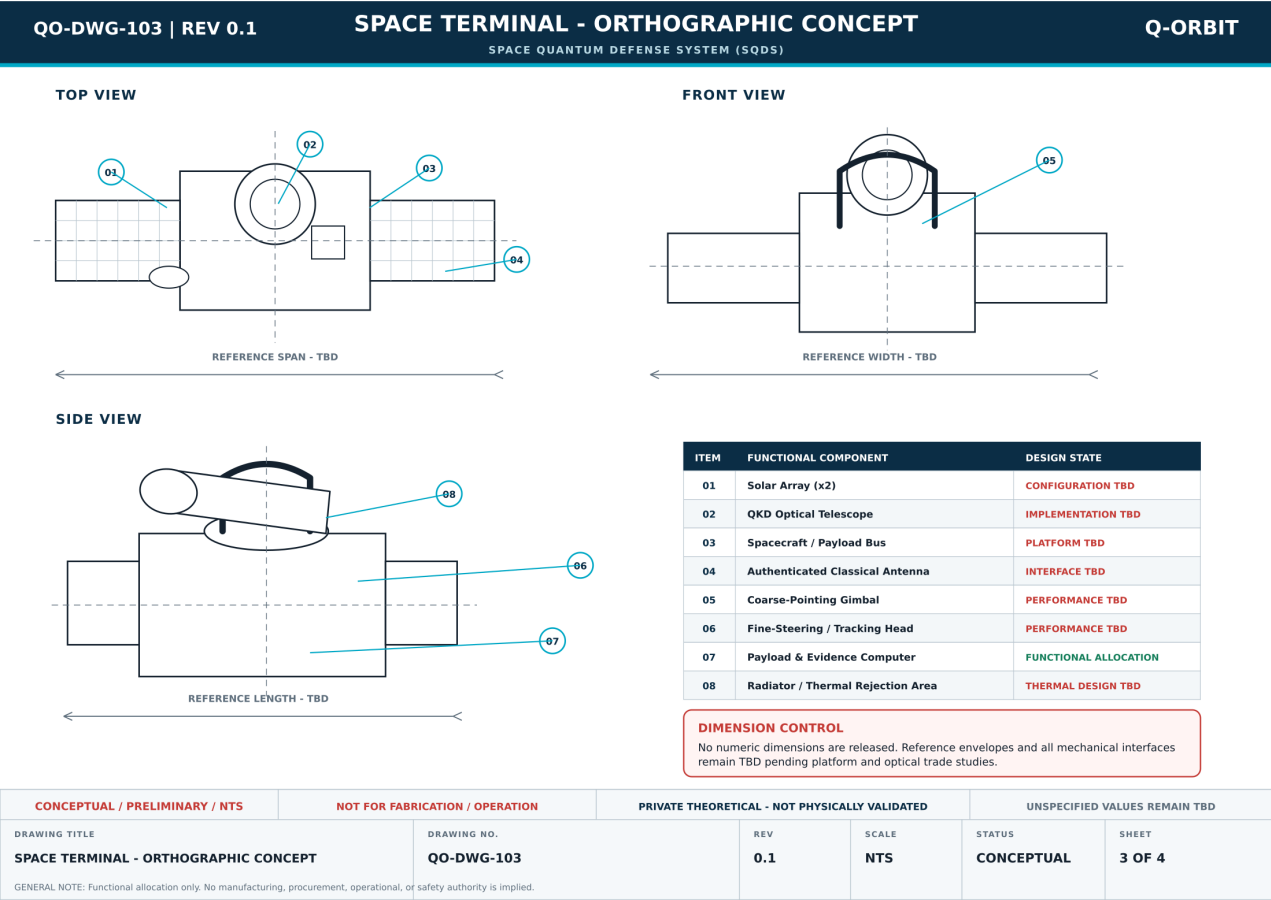


Figure 3: QO-DWG-103 Rev 0.1 - Space-terminal orthographic concept. Engineering decision / conceptual - not measured. No numeric mechanical dimensions are released; component envelopes, datums, tolerances, materials, mounting interfaces, and qualification requirements remain TBD. PRELIMINARY - NOT TO SCALE - NOT FOR FABRICATION - NOT PHYSICALLY VALIDATED.

5.4 End-to-End Link and Trust Boundary

Figure 4 connects the conceptual space segment, free-space optical path, ground segment, control plane, and key-management boundary. It preserves separation between optimization/control metadata and secret-key material.

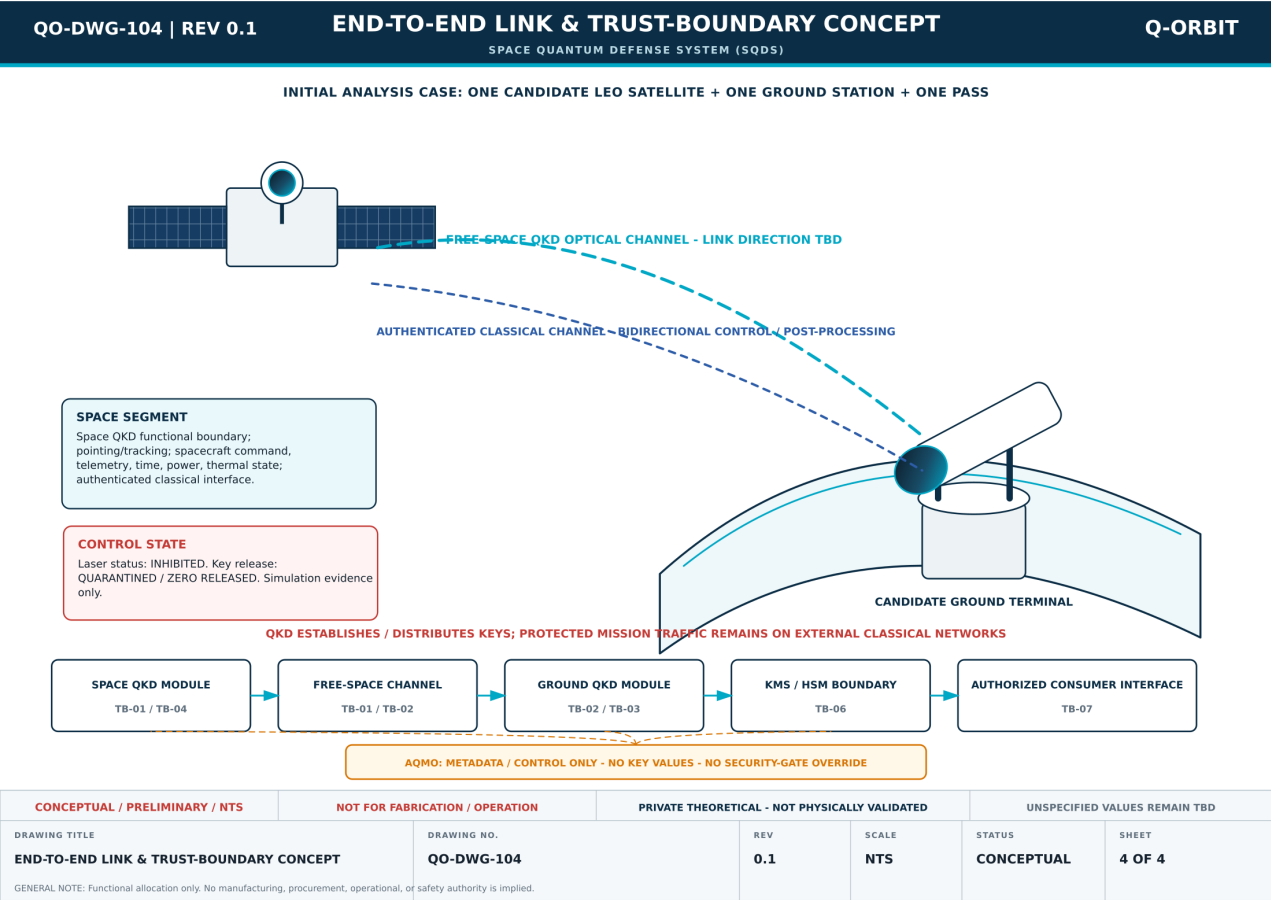


Figure 4: QO-DWG-104 Rev 0.1 - End-to-end link and trust-boundary concept. Engineering decision / conceptual - not measured. Optical link direction remains TBD. AQMO is limited to metadata and control, has no access to key values, and cannot override security gates. Key release remains QUARANTINED / ZERO RELEASED. PRELIMINARY - NOT TO SCALE - NOT PHYSICALLY VALIDATED.

The drawings do not close any proof, characterization, laser-safety, orbit, parcel, construction, or procurement gate. They make those unresolved interfaces visible and therefore support the same fail-closed evidence discipline as the numerical analysis.

6. Discussion

6.1 Positive Baseline versus Non-Robust Design Space

The exact positive baseline demonstrates deterministic reproduction, but the coupled screen shows that positivity is not robust across the declared research ranges. The negative median is especially important: the nominal point cannot be promoted to a mission claim without evidence-backed distributions and a validated physical model.

6.2 Why QBER Alone Is Insufficient

The finite-key expression also depends on vacuum and single-photon bounds, phase-error estimation, error-correction leakage, finite penalties, and the complete protocol profile. QBER cannot certify phase randomization, source leakage, detector mismatch, authentication, end-point behavior, or key-management state.

6.3 Model Response Is Not Device Specification

The one-parameter frontiers and local ranks freeze other variables and use arbitrary but declared steps. They are useful for finding fragile regions of the software fixture, but they are not tolerances, procurement filters, calibration limits, or security-proof domains.

6.4 Fail-Closed Evidence Handling

The absence of a numerical penalty for an unmapped effect is intentional. Assigning a convenient penalty would create an appearance of completeness without a proof basis. The fail-closed rule keeps the gap visible and prevents numerical output from widening the claim.

7. Limitations

- The reference loss curve and expected counts are modeled rather than measured.
 - Baseline reproduction does not provide an independent security proof.
 - Binary64 arithmetic is used; interval and arbitrary-precision sensitivity remain untested.
 - Only the half-window is globally searched; protocol probabilities and most intensities remain frozen.
 - The coupled ranges are engineering bounds rather than probability distributions.
 - No source, detector, atmosphere, parcel, orbit authority, hardware, or operational security evidence is used.
 - The study does not predict Tabuk availability, yield, or mission success.
 - All margins are upper bounds w.r.t. error-correction efficiency (frozen fixture assumes $f_{EC} = 1$; realistic $f_{EC} > 1$ would reduce margin).
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8. Conclusion

Q-Orbit demonstrates a reproducible theoretical workflow for finite-key satellite-QKD screening and makes the limitations of that workflow machine-checkable. The controlled fixture reproduces exactly, but the coupled screen is dominated by nonpositive cases and several device effects remain outside the selected proof representation. The scientifically defensible conclusion is not that a mission is ready, but that evidence and proof incompleteness can be exposed before they are mistaken for design maturity.

The next theoretical research task is a proof-profile selection and parameter-domain mapping for phase randomization, source flaws, pulse correlations, detector history, detector mismatch, and characterization confidence. Hardware procurement and field operation are neither required nor authorized for this submission.

Data and Reproducibility Statement

The controlled V0.16-TA1 package contains the frozen input index, executable model, generated CSV outputs, workbook, regression results, audits, and cryptographic hashes needed to reproduce the reported software results. Distribution remains private. Modeled candidate key material is not real key material and remains quarantined. Independent end-to-end re-execution is blocked pending controlled inputs REQ-01...03.

References

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